Robin Erb

Freelance Software Developer / M.Sc. Student in Data and Computer Science

Heidelberg, Germany | robin.erb2000@gmail.com | Portfolio | LinkedIn

PROFESSIONAL EXPERIENCE

campus.five Sep 2025 – Present

Freelance Software Developer · Remote

- **ERP and Shopify:** Built real-time middleware on Google Cloud Functions and Firestore to sync sales, returns, and inventory. Automated product drafting, publishing, and archiving.
- **Sales automation:** Implemented OCR for vehicle documents and helped deploy a chatbot to handle sales inquiries.
- **Dashboard:** Built a dashboard with automated file generation, WhatsApp integration, and tools to analyze sales and operational data.

University of Konstanz, HCI Department Sep 2023 – Mar 2025

Student Assistant · Konstanz, Germany

- Designed the SpatialMouse prototype in CAD and built its Arduino hardware, including C programming and circuit assembly.
- Built a virtual test environment for the prototype in Unity.

EDUCATION

Heidelberg University Apr 2025 – Jul 2027

M.Sc. Data and Computer Science · Heidelberg, Germany

Machine Learning specialization · Current grade: 1.4 (German grading scale).

Master's thesis: Exploring how reinforcement learning and hyperparameter optimization can make neural networks smaller and more efficient.

University of Copenhagen Feb 2026 – Jul 2026

Semester Abroad, M.Sc. Computer Science · Copenhagen, Denmark

Selected coursework: Online and Reinforcement Learning, Web Recommender Systems.

University of Konstanz Oct 2021 – Mar 2025

B.Sc. Computer Science · Konstanz, Germany

Data Science and Human-Computer Interaction · Final grade: 1.4 (German grading scale).

Bachelor's thesis (grade: 1.0): Developed SpatialMouse for 2D/3D interaction, conducted a user study, and analyzed the results in Python.

PUBLICATIONS

S. Hubenschmid, J. Zagermann, **R. Erb**, et al. SpatialMouse: A Hybrid Pointing Device for Seamless Interaction Across 2D and 3D Spaces. *ACM VRST '25*. DOI: 10.1145/3756884.3766047.

ADDITIONAL EXPERIENCE AND VOLUNTEERING

Bouldering Instructor, Steinbock, Konstanz Sep 2021 – Jul 2023

Federal Voluntary Service (BFD), Kletterwerk Radolfzell Sep 2020 – Aug 2021

Taught bouldering and rope climbing to school groups.

TECHNICAL SKILLS AND LANGUAGES

Programming: Python, Java, C, C++, SQL, R
ML and data: PyTorch, scikit-learn, pandas, OCR
Agentic coding: OpenAI Codex, Claude Code, Antigravity
App development: Vercel, Supabase, Cloudflare
Cloud and tools: Google Cloud Functions, Firestore, Git, Docker, Unity, CAD
Languages: German (native), English (fluent), French (basic)